SMART LIBRARY MANAGEMENT SYSTEM

Alloy Formal Specification

1. Introduction:

The Smart Library Management System is formally verified using Alloy Analyzer.

Alloy is used to model relationships, verify constraints, and identify inconsistencies through counterexample analysis.

This verification ensures correctness of book issuing rules, borrowing constraints, and system consistency.

2. Core Features:

The system includes the following functionalities:

- Book Issuing
- Book Returning
- Book Availability Checking
- Maximum 3 Books Constraint
- One Book Assigned to One Student
- Fine Management
- Barcode Validation
- Notifications and Reminders

3. Relational Model Definition:

Alloy Specification:

```alloy
sig Student {}

sig Book {}

one sig Library {
    issued : Book -> lone Student
}
```

Explanation:

This relational model defines:

- Student → registered students
- Book → books in the library
- issued → relationship between books and students

The relation ensures that a book can only be mapped to one student.

4. Maximum 3 Books Constraint:

Alloy Specification:

```alloy
fact MaxThreeBooksRule {
    all s : Student |
        #(Library.issued.s) <= 3
}
```

Explanation:

This constraint ensures:

- A student can issue a maximum of three books.

The system prevents invalid states where a student exceeds the borrowing limit.

Formal rule:

$$\forall s: STUDENT \cdot \#(issuedBooks(s)) \leq 3$$

Screenshot:

C:\Users\Zainny\Desktop\library.als

File Edit Execute Options Window Help

New Open Reload Save Execute Show

```
sig Student {}

sig Book {}

one sig Library {
   issued : Book -> lone Student
}

fact MaxThreeBooksRule {
   all s : Student |
      #(Library.issued.s) <= 3
}

fact OneBookOneStudent {
   all b : Book |
      lone (Library.issued[b])
}

assert NoStudentHasMoreThan3Books {
   all s : Student |
      #(Library.issued.s) <= 3
}

assert NoBookIssuedToMultipleStudents {
   all b : Book |
      lone (Library.issued[b])
}

check NoStudentHasMoreThan3Books for 5
check NoBookIssuedToMultipleStudents for 5
```

```
Executing "Check NoStudentHasMoreThan3Books for 5"
   Solver=sat4j Bitwidth=0 MaxSeq=0 SkolemDepth=1 Symmetry=20
   603 vars. 40 primary vars. 1152 clauses. 75ms.
   No counterexample found. Assertion may be valid. 18ms.
```

5. One Book → One Student Constraint:

Alloy Specification:

```
alloy

fact OneBookOneStudent {
    all b : Book |
        lone (Library.issued[b])
}
```

Explanation:

This rule guarantees:

- One book cannot be assigned to multiple students simultaneously.

This maintains consistency and prevents duplicate issuing.

6. Assertion Verification:

Alloy Specification:

```
alloy

assert NoStudentHasMoreThan3Books {
    all s : Student |
        #(Library.issued.s) <= 3
}

assert NoBookIssuedToMultipleStudents {
    all b : Book |
        lone (Library.issued[b])
}
```

Explanation:

Assertions are used to verify correctness of system behavior.

The system checks:

- No student exceeds the maximum borrowing limit
- No book is assigned to multiple students

7. Model Checking:

Alloy Specification:

```alloy
check NoStudentHasMoreThan3Books for 5
check NoBookIssuedToMultipleStudents for 5
```

Explanation:

The check command verifies assertions within a finite scope.

In this model:

- Alloy checks correctness using up to 5 instances

If no violation exists:

```
No counterexample found
```

This confirms system correctness.

8. Counterexample Analysis (Mandatory):

Modified Alloy Specification:

To test system robustness, the borrowing limit constraint was temporarily removed.

Removed constraint:

```alloy
fact MaxThreeBooksRule {
    all s : Student |
        #(Library.issued.s) <= 3
}
```

Explanation:

After removing the constraint:

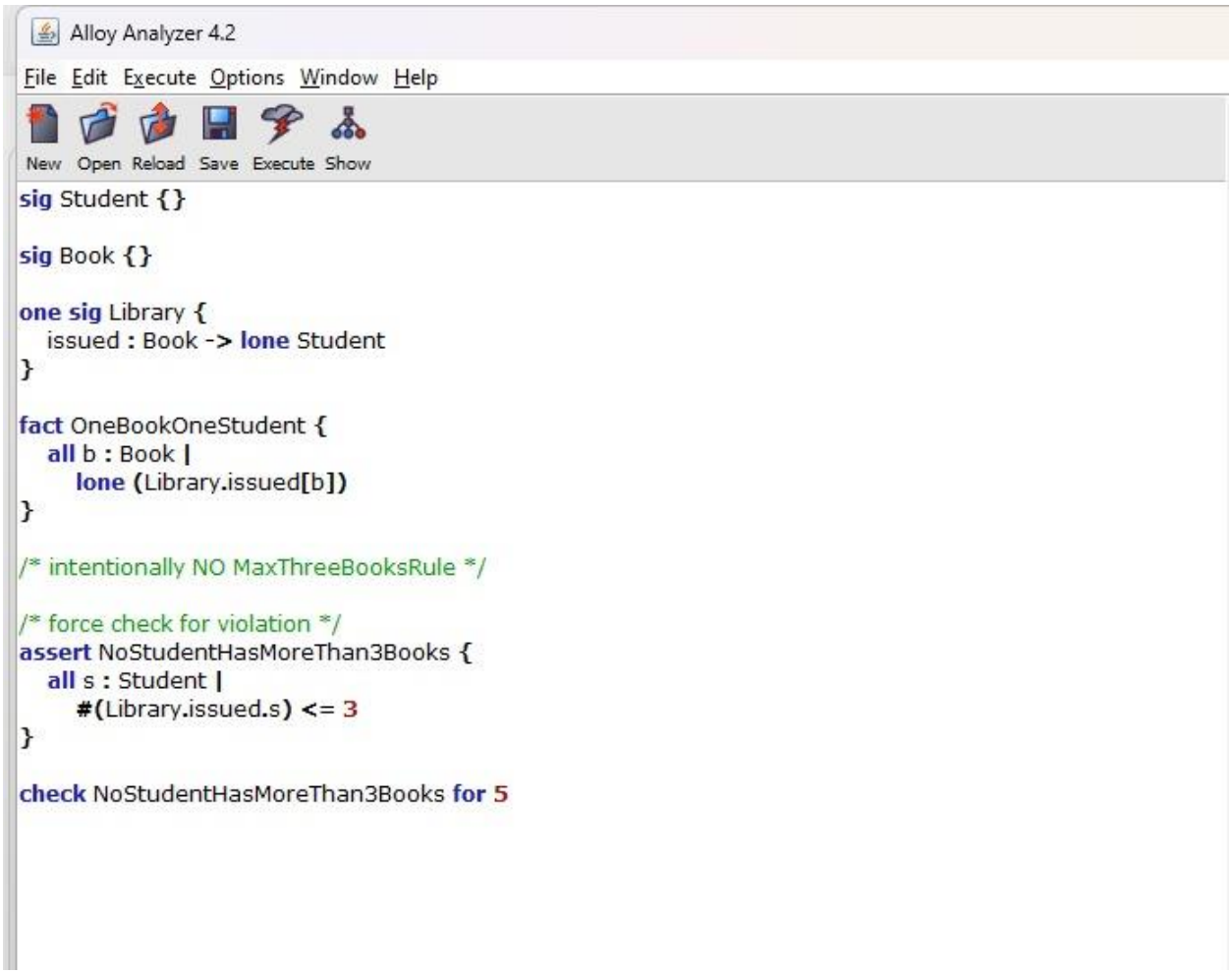
- Alloy generated a counterexample

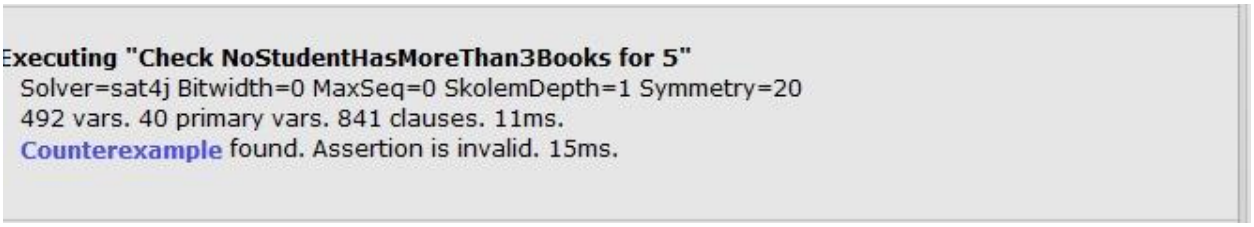
The counterexample showed:

- A student receiving more than 3 books

This proves that the invariant is necessary for system correctness.

Screenshot:



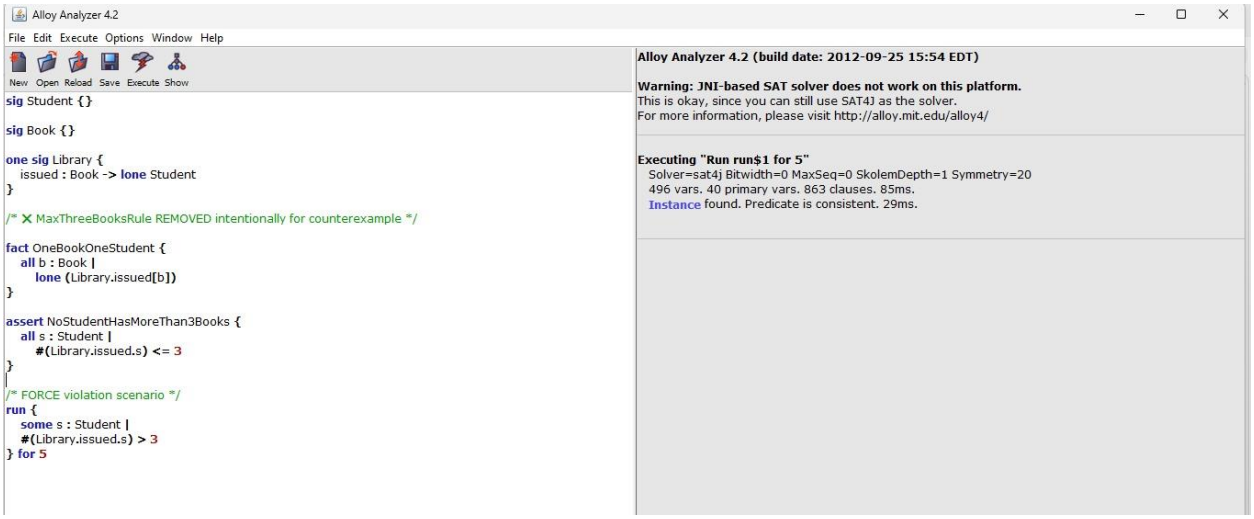


9. Verification Result:

The Alloy model successfully verified:

- Borrowing constraints
- Unique book assignment
- Relational correctness
- Constraint enforcement

Verification also demonstrated that invalid states are detected through counterexamples.



10. Conclusion:

The Smart Library Management System was successfully verified using Alloy Analyzer.

The verification process confirmed:

- ✓ Logical correctness
- ✓ Relational consistency
- ✓ Borrowing rule enforcement
- ✓ Detection of invalid states using counterexamples

Alloy provided strong structural verification for the system.